SCHOOL OF COMPUTING SCIENCES

Pak-Austria Fachhochschule: Institute of Applied Sciences and Technology

Haripur, Pakistan

Customer Shopping Behavior Analysis

Data Visualization Lab (COMP-434L)

6th Semester — BS(DS) – F23

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3. Abstract

This project presents an analytical investigation into customer shopping behavior using transactional data comprising 3,900 purchase records across various product categories. The primary objective is to uncover meaningful insights into spending patterns, customer demographics, product preferences, subscription behavior, and discount utilization in order to support strategic business decision-making. The dataset, consisting of 18 features including customer demographics, purchase details, and behavioral indicators, was processed and cleaned using Python, with exploratory data analysis conducted via the pandas library. Key preparation steps included handling 37 missing values in the Review Rating column through category-wise median imputation, feature engineering to derive age groups and purchase frequency metrics, and integration with a PostgreSQL database for structured SQL-based analysis. Ten targeted SQL queries were executed to extract insights such as revenue by gender, top-rated products, subscription versus non-subscription spending, and customer segmentation into New, Returning, and Loyal categories. The findings reveal that male customers contributed significantly more revenue than female customers, young adults were the highest-spending age group, and the vast majority of customers (73%) were non-subscribers despite having similar average spend per transaction. A comprehensive interactive dashboard was built in Power BI to visualize these findings, incorporating KPI cards, donut charts, bar charts, a geographic map, and gender-based purchase breakdowns. The project concludes with actionable business recommendations including targeted subscription conversion strategies, gender-specific marketing campaigns, and cross-sell opportunities for weaker product categories.

4. Problem Statement

4a. What Business/Problem Are You Visualizing?

Retail businesses today collect vast amounts of transactional data but often struggle to extract actionable patterns from it. This project focuses on a customer shopping behavior dataset that captures real-world purchase activity. The core problem being visualized is understanding how different customer segments — defined by gender, age group, subscription status, and purchase frequency — contribute to overall revenue, and how product preferences, discount usage, and shipping choices vary across these segments. Without effective visualization, such multi-dimensional data remains difficult to interpret for business stakeholders.

4b. Objective of the Dashboard/Report

- To analyze and visualize total revenue distribution across gender, age groups, and subscription status.
- To identify top-performing and discount-dependent products across all categories.
- To segment customers into behavioral groups (New, Returning, Loyal) to guide retention strategies.
- To compare shipping preferences and their relationship with purchase amounts.
- To provide an interactive Power BI dashboard enabling dynamic filtering by gender and subscription status.

4c. Real-World Use of This Project

The insights derived from this project have direct applications in the retail and e-commerce sectors. Marketing teams can use gender- and age-based revenue data to design targeted advertising campaigns. Supply chain managers can optimize inventory by focusing on high-demand product categories such as Clothing. Customer relationship managers can use segmentation data to design loyalty programs aimed at converting non-subscribers and retaining loyal customers. The Power BI dashboard can be embedded in business intelligence platforms to support real-time decision-making by executives and analysts.

5. Dataset Description

5a. Dataset Source

The dataset used in this project is a publicly available customer shopping behavior dataset sourced from Kaggle. It simulates real-world retail transaction data and is widely used for data analysis and machine learning practice.

5b. Number of Records and Fields

Total Records (Rows): 3,900

Total Features (Columns): 18

5c. Data Types of Columns

Column Name	Data Type	Description
Customer ID	Integer	Unique identifier for each customer
Age	Integer	Customer age in years
Gender	Text	Customer gender (Male/Female)
Item Purchased	Text	Name of the product bought
Category	Text	Product category (Clothing, Accessories, etc.)
Purchase Amount (USD)	Float	Transaction value in US dollars
Location	Text	US state of the customer
Size	Text	Product size (S/M/L/XL)
Color	Text	Product color
Season	Text	Season of purchase
Review Rating	Float	Customer rating (1.0 to 5.0)
Subscription Status	Text	Whether customer has a subscription
Shipping Type	Text	Shipping method selected
Discount Applied	Text	Whether a discount was applied

Column Name	Data Type	Description
Promo Code Used	Text	Whether a promo code was used
Previous Purchases	Integer	Number of prior purchases
Payment Method	Text	Method of payment used
Frequency of Purchases	Text	How often the customer shops

5d. Screenshot of 5 Sample Rows

The table below shows five representative sample rows from the dataset to illustrate the structure and nature of the data fields:

Cust. ID	Age	Gender	Item	Category	Amount ($)	Season	Rating	Subscription
1	55	Male	Blouse	Clothing	53	Spring	4.2	No
2	19	Female	Sneakers	Footwear	64	Summer	3.9	No
3	44	Male	Jacket	Outerwear	73	Fall	4.5	Yes
4	32	Female	Jewelry	Accessories	90	Winter	3.5	No
5	28	Male	Hat	Accessories	97	Spring	4.8	Yes

6. Data Preparation

The following data preparation steps were performed in Python using the pandas library before loading the data into PostgreSQL for SQL analysis.

6a. Data Cleaning Steps Performed

- Imported the dataset using pd.read_csv() and performed an initial structural inspection using df.info() and df.describe() to understand data types, column ranges, and distribution of values.
- Verified all 3,900 rows were correctly loaded with 18 columns intact.
- Checked for duplicate Customer IDs and confirmed each record was unique.

6b. Handling Missing Values

A total of 37 missing values were found exclusively in the Review Rating column. Rather than removing these records (which would reduce the dataset), the missing values were imputed using the median review rating computed per product category. This approach preserves data volume while ensuring statistical consistency within each category.

6c. Data Transformation

- Column Standardization: All column names were renamed to snake_case (e.g., Purchase Amount (USD) became purchase_amount_usd) for consistency and ease of use in SQL queries.
- Feature Engineering — Age Group: A new age_group column was created by binning the Age column into four groups: Young Adult (18–30), Adult (31–45), Middle-Aged (46–60), and Senior (61+).
- Feature Engineering — Purchase Frequency Days: A purchase_frequency_days column was derived from the Frequency of Purchases column to convert categorical frequency descriptions into approximate numeric intervals.
- Redundancy Check: The promo_code_used column was found to be perfectly correlated with the discount_applied column and was therefore dropped to eliminate redundancy.

6d. Relationships Created Between Tables

No multi-table relational model was required for this project as the dataset is structured as a single flat table. However, the cleaned DataFrame was exported to a PostgreSQL database as the shopping_behavior table. Derived query results (such as customer segments and revenue by age group) were treated as virtual tables through SQL views and subqueries.

6e. Data Model / Schema View

The data model consists of a single primary table — shopping_behavior — with the following key schema structure after cleaning and transformation:

Column Name	Data Type	Constraints
customer_id	BIGINT	Primary Key
age	INTEGER	NOT NULL
gender	TEXT	NOT NULL
item_purchased	TEXT	NOT NULL
category	TEXT	NOT NULL
purchase_amount_usd	NUMERIC	NOT NULL
location	TEXT	-
review_rating	NUMERIC	Imputed (no NULLs post-cleaning)
subscription_status	TEXT	NOT NULL
discount_applied	TEXT	NOT NULL
previous_purchases	INTEGER	NOT NULL
age_group	TEXT	Derived column

7. Dashboard Design Planning

7a. KPIs and Metrics Selected

The following key performance indicators (KPIs) were selected to provide a concise, high-level overview of the dataset at the top of the dashboard:

KPI / Metric	Value	Reason for Inclusion
Total Customer Count	3,900	Gives an immediate scale of the dataset
Average Purchase Amount	$59.76	Establishes a baseline for spending analysis
Average Review Rating	3.75 / 5.0	Indicates overall customer satisfaction
Average Customer Age	44 years	Provides demographic context for targeting

7b. Layout Planning of Dashboard

The dashboard was designed with a dark-themed aesthetic to improve readability on screens and to give a professional, modern appearance. The layout follows a top-to-bottom hierarchy:

- Row 1 (Top): Four KPI cards displaying key summary statistics.
- Row 2 (Middle): Three visualizations — a horizontal bar chart (purchase by category), a donut chart (subscription status breakdown), and a geographic map (market dominance by location).
- Row 3 (Bottom): Two additional charts — a horizontal bar chart (revenue by age group) and a pie chart (purchase split by gender).
- Right Panel: Two interactive slicers — one for Gender filter (Female/Male) and one for Subscription filter (Yes/No) — allowing dynamic cross-filtering of all visuals.

7c. Type of Charts Selected and Reason

Chart Type	Used For	Reason
KPI Card	Summary metrics	Instant snapshot of key numbers
Horizontal Bar Chart	Purchase amount by category	Easy comparison across 4 categories
Donut Chart	Subscription status split	Shows proportions clearly (27% vs 73%)

Chart Type	Used For	Reason
Geographic Map	Market dominance by location	Spatial distribution of customers
Horizontal Bar Chart	Revenue by age group	Ranked comparison across age segments
Pie Chart	Purchase split by gender	Simple 2-way proportion visualization

8. Visualizations Created

The Power BI dashboard included the following key visualizations, each designed to answer a specific business question:

8.1 Total Purchase Amount by Category

A horizontal bar chart was created to compare total purchase amounts across the four product categories: Clothing, Accessories, Footwear, and Outerwear. Clothing emerged as the dominant category, followed by Accessories. This visualization helps identify which product lines drive the most revenue.

8.2 Customer Count by Subscription Status

A donut chart was used to visualize the proportion of subscribers versus non-subscribers. The chart clearly shows that 2,847 customers (73%) have no subscription, while only 1,053 (27%) are active subscribers. This insight is critical for customer retention and subscription growth strategies.

8.3 Market Dominance Map

A geographic map visual was included to show the distribution of customer purchases across different US states. Darker regions represent higher transaction volumes, with notable market concentration in certain coastal and midwestern states.

8.4 Sum of Purchase Amount by Age Group

A horizontal bar chart broke down total spending by age group. The Young Adult segment (18–30 years) was the highest spender at $62,143, followed closely by the Middle-Aged group at $59,197. This chart informs age-targeted marketing strategies.

8.5 Purchase by Gender

A pie chart compared total purchase amounts between male and female customers. Male customers accounted for 68% of total purchases ($157,890) versus 32% for female customers ($75,191), indicating a significantly male-dominated customer base.

8.6 Interactive Slicers

Two slicers — one for Gender and one for Subscription Status — were added to enable dynamic filtering of all dashboard visuals simultaneously. This allows stakeholders to drill down into specific segments without rebuilding any reports.

9. Calculated Fields / Measures

9a. List of Calculated Fields / DAX Measures

Measure Name	Formula / Logic	Purpose
Total Revenue	SUM(purchase_amount_usd)	Aggregate total revenue across all filters
Average Purchase Amount	AVERAGE(purchase_amount_usd)	Shows mean spend per transaction
Average Review Rating	AVERAGE(review_rating)	Tracks average product satisfaction
Customer Count	COUNT(customer_id)	Number of customers in the current filter context
Average Customer Age	AVERAGE(age)	Mean age for demographic KPI card
Revenue by Gender	CALCULATE(SUM, FILTER by gender)	Used for gender pie chart
Subscriber Revenue %	DIVIDE(sub_revenue, total_revenue)	Subscription contribution ratio
Age Group Revenue	CALCULATE(SUM, FILTER by age_group)	Revenue per age segment

9b. Purpose of Each Calculation

- Total Revenue is the foundational measure driving all revenue-related visuals and KPI cards.
- Average Purchase Amount provides the per-transaction baseline for comparing customer segments.
- Average Review Rating reflects customer satisfaction and supports product quality decisions.
- Customer Count enables demographic analysis and segment sizing.
- Revenue by Gender feeds the gender-based pie chart and supports gender-targeted marketing analysis.
- Subscriber Revenue % helps quantify the monetization gap between subscribers and non-subscribers.
- Age Group Revenue enables the age-based bar chart and informs targeting strategy by life stage.

9c. SQL Query Examples (Business Transactions)

The following SQL queries were executed in PostgreSQL as part of the structured business analysis:

```sql
-- Revenue by Gender
SELECT gender, SUM(purchase_amount_usd) AS revenue FROM
shopping_behavior GROUP BY gender;
```

```sql
-- Top 5 Products by Average Rating
SELECT item_purchased, ROUND(AVG(review_rating),2) AS avg_rating FROM
shopping_behavior GROUP BY item_purchased ORDER BY avg_rating DESC
LIMIT 5;
```

10. Insights from Dashboard

10a. Key Insights Discovered from the Visualization

- Male customers generated $157,890 in total revenue (68%) compared to $75,191 from female customers (32%), confirming a highly male-dominated customer base.
- Young Adults (ages 18–30) are the top revenue-contributing age group at $62,143, closely followed by Middle-Aged customers at $59,197.
- Only 27% of customers hold an active subscription despite both groups showing nearly identical average spend ($59.49 vs $59.87), indicating a significant untapped subscription growth opportunity.
- Gloves, Sandals, and Boots are the top three highest-rated products with ratings of 3.86, 3.84, and 3.82 respectively.
- Express shipping customers spend slightly more on average ($60.48) compared to Standard shipping customers ($58.46), suggesting express users may be higher-value customers.

10b. Patterns and Trends Observed

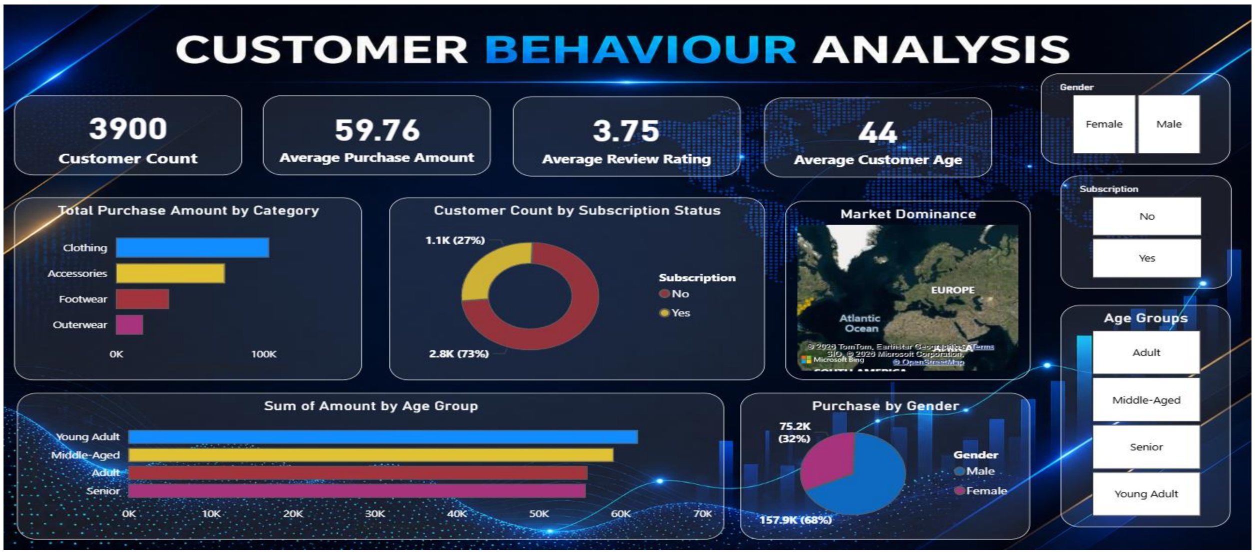
- Hat, Sneakers, and Coat show the highest discount dependency (50%, 49.66%, 49.07%), suggesting these products may rely on promotions to drive purchases rather than organic demand.
- Clothing dominates all other categories in total purchase volume, with Blouse, Pants, and Shirt being the top three most purchased clothing items.
- The vast majority of customers (3,116 out of 3,900) fall into the Loyal segment, suggesting strong baseline retention — but this also means relatively few new customers are being acquired (only 83 classified as New).
- Repeat buyers (previous purchases > 5) overwhelmingly do not subscribe (2,518 vs 958), which is counterintuitive and represents a key strategic gap.

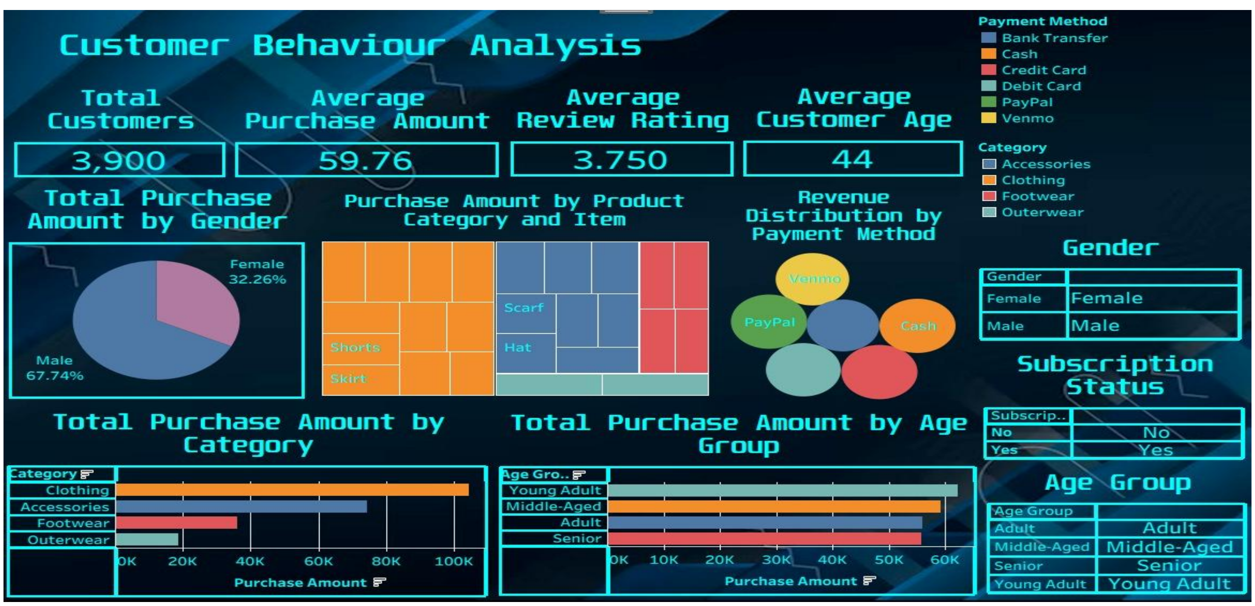
10c. Business Decisions That Can Be Made

- Launch targeted subscription campaigns directed at male customers aged 18–45 who are repeat buyers but have not yet subscribed, as this group represents the highest revenue potential.
- Reduce blanket discounts on Hat, Sneakers, and Coat and instead trial value-based pricing or bundle offers to improve margin on these discount-dependent products.
- Invest in female-targeted marketing campaigns for Accessories and Footwear, which represent categories with room for growth in the female segment.
- Develop an Express Shipping loyalty perk to attract more of the higher-spending express-preference customer segment.

11. Final Dashboard Output

The final interactive dashboard was built in Microsoft Power BI Desktop. The screenshot below represents the completed dashboard output, displaying all visualizations, KPI cards, and interactive slicers on a single dark-themed canvas:





The dashboard features the following components on a single page:

- Four KPI summary cards at the top: Customer Count (3,900), Average Purchase ($59.76), Average Rating (3.75), and Average Age (44).
- Total Purchase Amount by Category — horizontal bar chart.
- Customer Count by Subscription Status — donut chart showing 73% non-subscribers.
- Market Dominance — filled geographic map of North America.
- Sum of Purchase Amount by Age Group — horizontal bar chart.
- Purchase by Gender — pie chart (68% male, 32% female).
- Two interactive slicers for Gender and Subscription Status filtering.

12. Challenges Faced

- Missing Data Imputation: The 37 missing values in the Review Rating column required careful handling. A simple global mean imputation would have distorted product-specific ratings, so category-wise median imputation was implemented, which required groupby operations and conditional logic in Python.
- Data Redundancy Detection: Identifying that promo_code_used and discount_applied were completely redundant required verifying their correlation across all 3,900 rows before safely dropping the column without loss of information.
- PostgreSQL Integration: Setting up the SQLAlchemy connection string and ensuring the pandas DataFrame schema matched the PostgreSQL table types required debugging of data type mismatches, particularly for boolean columns.
- Power BI Performance: Loading a 3,900-row dataset with 17 columns into Power BI with multiple calculated measures and slicers initially caused refresh delays, which were resolved by optimizing DAX measure logic and reducing redundant calculated columns.
- Geographic Map Visualization: Mapping customer locations (US state names) to the Power BI map visual required enabling Bing Maps integration and ensuring location data was classified as the correct geographic type in the data model.
- Dashboard Theme Consistency: Applying a uniform dark theme across all visuals while maintaining readability of text and chart labels required multiple iterations of color calibration and font sizing.

13. Conclusion

This project successfully demonstrated an end-to-end data visualization workflow applied to a real-world retail shopping behavior dataset. Beginning with data ingestion and cleaning in Python, the project progressed through exploratory analysis, SQL-based business transaction querying in PostgreSQL, and culminated in the development of a comprehensive interactive dashboard in Power BI. The analysis revealed several significant findings: male customers dominate purchase volumes, young adults are the highest-spending demographic, and a large majority of high-frequency buyers remain unsubscribed despite comparable per-transaction spending. These insights, when translated into actionable business recommendations, have the potential to improve subscription conversion rates meaningfully, optimize marketing spend by demographic segment, and increase average order values through targeted upselling and bundling strategies. The project also provided practical experience with multi-tool data workflows, reinforcing competencies in Python (pandas), SQL (PostgreSQL), and business intelligence visualization (Power BI). The interactive dashboard we produced is a functional prototype that could be deployed in a real retail analytics environment to support ongoing decision-making by business stakeholders.